Backward Design Lesson Plan:
Computational Biophysics & Rational Drug Design

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Course: Honors Biotechnology
School: Walter Payton College Preparatory High School

Stage 1: Desired Results

Established Goals

- Students will apply principles of structural biology, molecular thermodynamics, and pharmacokinetics to discover or optimize a small-molecule drug candidate targeting a specific disease protein.
- Students will navigate and extract data from professional, open-access bioinformatics databases and predictive modeling tools (RCSB PDB, SwissDock, SwissADME) without requiring command-line programming.

Essential Questions (EQs)

1. How do structural configurations at the atomic level dictate macro-level physiological health and disease?
2. Why is a high chemical affinity (binding strength) alone insufficient to make a molecule a viable medical drug?
3. How do modern computational scientists balance structural biochemistry with systemic physiology when engineering therapeutics?

Key Understandings

Students will understand that...

- **Structure Dictates Function:** Identifying the specific three-dimensional topography and active binding pockets of a protein is a prerequisite to therapeutic intervention.
- **Thermodynamic Favorability:** The binding affinity between a ligand and a receptor is represented by Gibbs Free Energy (Δ Delta G), where increasingly negative values indicate stronger, more spontaneous interactions.
- **Systemic Viability:** A drug candidate must be able to survive the human body's metabolic and physical barriers (absorption, blood-brain barrier, molecular weight constraints) to reach its target safely.

Stage 2: Assessment Evidence

Performance Tasks (Summoning the Capstone)

- **Group Oral Presentation & Professional Defense:** A comprehensive group pitch simulating a biophysics research symposium. Teams must justify their target choice, present clean 3D renderings of their protein-ligand interfaces, present comparative matrices of their thermodynamic docking scores, and evaluate the physiological ADME viability of their top candidate.
- **Peer & Instructor Q&A Defense:** Students must individually field technical questions regarding their molecular selections, interpretation of Delta G, and Lipinski parameters.

Formative Benchmarks (Checking for Understanding)

- **Milestone 1:** PDB ID submission accompanied by a verified structural rendering highlighting the target protein's native ligand and active site coordinates.
- **Milestone 2:** SwissDock data spreadsheet containing calculated Delta G scores for 2–3 test compounds compared strictly against a designated positive control.
- **Milestone 3:** An ADME comparison sheet highlighting any structural or barrier violations according to Lipinski's Rule of Five.

Stage 3: Learning Plan (W.H.E.R.E.T.O. Framework)

Phase 0: Preliminary instruction on the following topics to prepare students for the capstone project (Approx. 4 Days)

- Functional groups
- Amino acids, peptides, and proteins
- Protein structure & function
- Cellular communication (e.g., ligands, receptors, delta Gibbs measurement)
- Cell signaling (e.g., GPCR)

Phase 1: Target Selection & Molecular Visualization (Approx. 2 Days)

- **Hook & Direct Instruction:** Introduce the concepts of structural biology using your own research paradigm (e.g., how cancer cells utilize chaperone proteins like CD147/MCT4 to alter their microenvironment) to illustrate why targeting protein architecture matters.
- **Guided Practice:** Walk students through the **RCSB Protein Data Bank (PDB)**. Demonstrate how to search for a disease target, isolate the structural coordinates, and utilize built-in or basic web-based visualization tools to view surface topography, find active binding pockets, and locate the native ligand.
- For example: See image 1 for an example of a protein called Human COX-1 Crystal Structure.

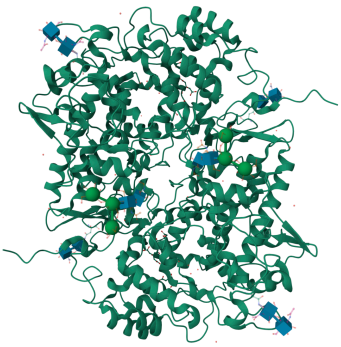


Image 1. Human COX-1 Crystal Structure

Key Questions:

  - What disease is the target?
  - Identify the key protein associated with this target disease and explain its physiological function or role in the disease.
- **Student Activity:** Students can look at a protein *without* its native ligand, use PrankWeb to identify the top three predicted pockets, and use the visualizer to see which amino acids line those pockets. This is a fantastic way to explicitly compare a known active site with alternative, potential allosteric sites.
- **Collaborative Work:** In small research groups, students select a specific disease target, retrieve its structural "blueprint" (.pdb file), and take high-resolution, labeled screenshots identifying the binding site. The groups are to select one positive control ligand and two experimental ligands.
  - Using molecular visualization software, they map the protein's surface topography, identify key active or allosteric binding pockets, and isolate the native ligand.
  - **PrankWeb (Web-Based / No Installation)**
  - Best for: Automatically locating and visualizing active and allosteric pockets instantly.
  - **How it works:** PrankWeb uses a machine-learning algorithm called P2Rank. Instead of making students manually hunt for a pocket, they simply type in a PDB ID code. The tool instantly predicts and color-codes the top potential ligand-binding pockets based on the protein's physical layout
  - For example: See image 2 for the Human COX-1 crystal structure analyzed by PrankWeb for active and allosteric sites.

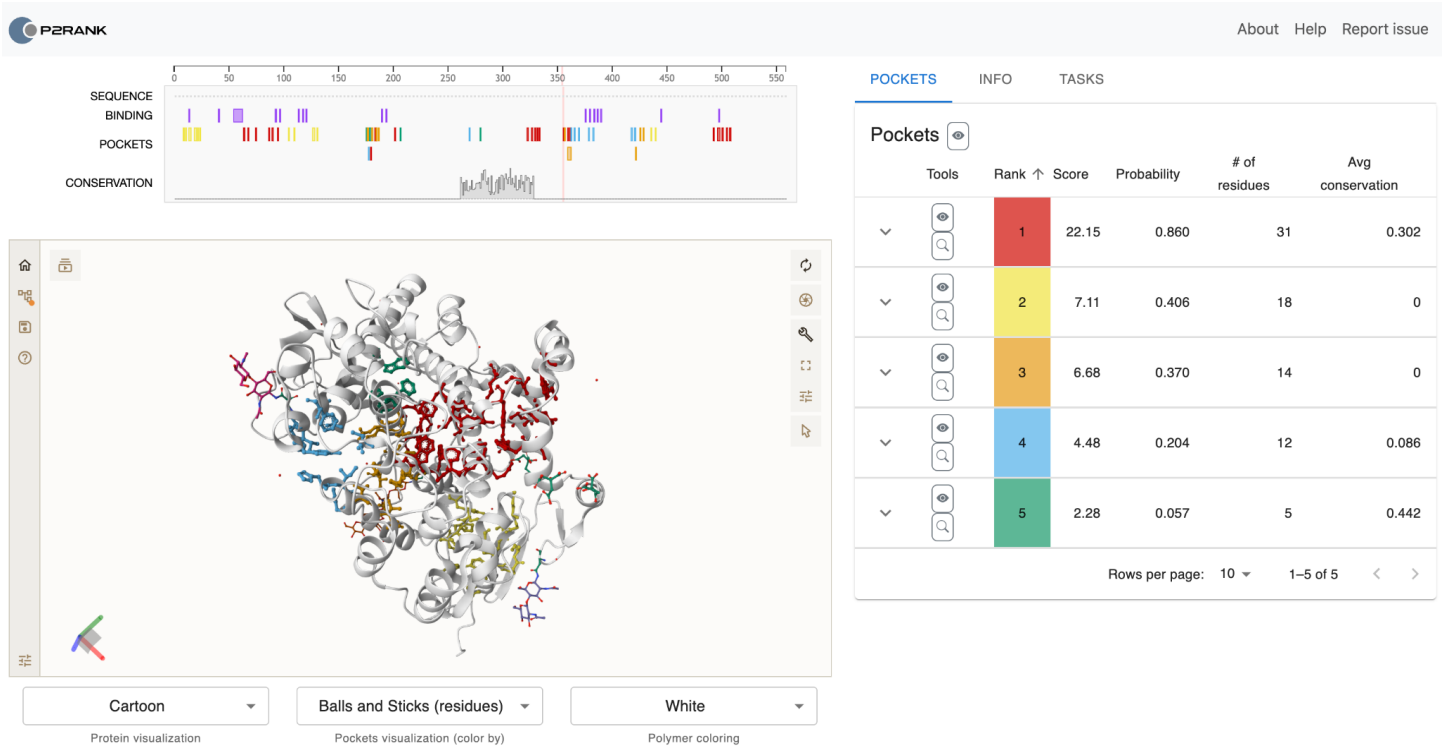
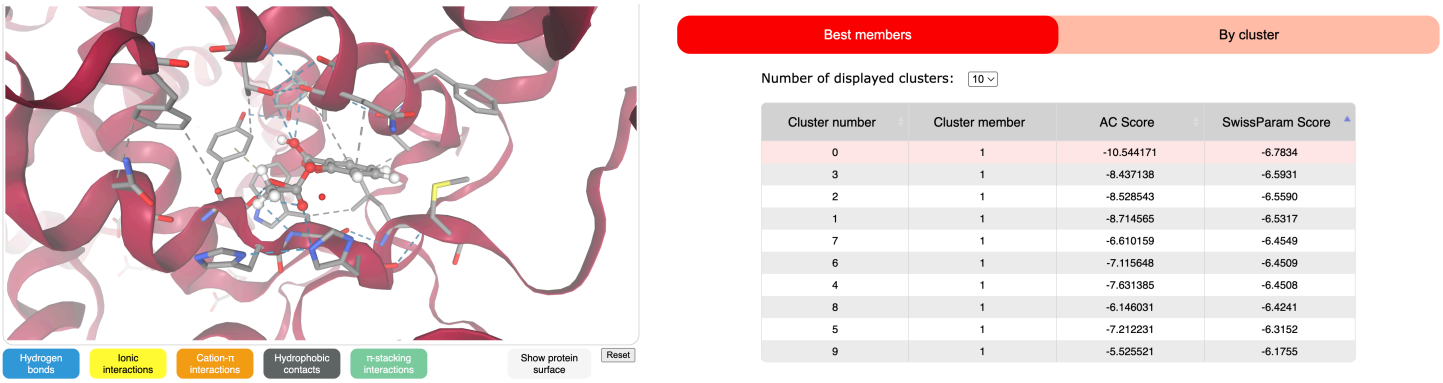


Image 2. Human COX-1 Crystal Structure Analyzed for Active and Allosteric Sites Using PrankWeb

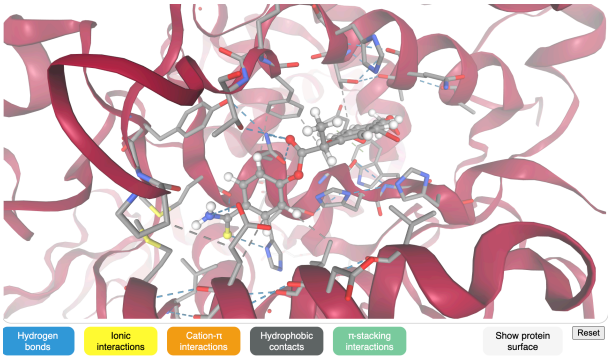
Phase 2: Virtual Screening & Molecular Docking (Approx. 1-2 Days)

- **Direct Instruction:** Introduce the thermodynamics of molecular docking. Define Gibbs Free Energy (Delta G) and explain the standard affinity scale:
  - *Weak:* approx -6 kcal/mol or less
  - *Moderate:* -6 to -10 kcal/mol
  - *Strong:* < -10 kcal/mol
- **Guided Practice:** Demonstrate how to input a protein target and a candidate small-molecule ligand into **SwissDock**. Show how to read the output files to extract predicted binding affinities (kcal/mol).
- For example: Aspirin and Human COX-1 Protein predicted binding affinities were analyzed.



Images 3. Aspirin and Human COX-1 Protein predicted binding affinities were analyzed by SwissDock.

- For example: An experimental drug called Otenaproxesul (ATB-346) (or Nitroaspirin / NCX-4016) & Human COX-1 Protein.



Best members | By cluster

Number of displayed clusters: 10

Cluster number	Cluster member	AC Score	SwissParam Score
5	1	24.616581	-8.4722
1	1	19.388252	-8.3597
6	1	24.689744	-8.2731
0	1	17.709214	-8.1740
8	1	24.985519	-7.9278
3	1	23.208544	-7.9082
2	1	21.048912	-7.7225
4	1	23.794140	-7.6851
9	1	25.161682	-7.6638
7	1	24.749638	-7.2924

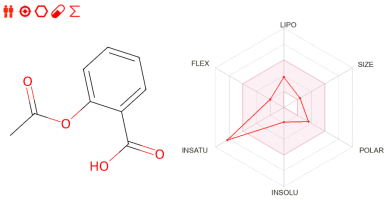
Images 4. ATB-346 and Human COX-1 Protein predicted binding affinities were analyzed by SwissDock.

- **Independent Group Work:** Groups select 2–3 chemical compounds to screen against their chosen protein pocket. These compound files can be found and downloaded from the **PubChem** website. They *must* identify and dock a known native ligand or existing pharmaceutical drug to serve as their baseline positive control. Groups compile these values into a matrix comparing the thermodynamic stability of each candidate.

Phase 3: Drug Delivery & ADME Profiling (Approx. 1-2 Days)

- **Direct Instruction:** Shift the focus from chemical affinity to physiological reality. Introduce the fundamentals of pharmacokinetics, focusing on the absolute necessity of systemic transport: Gastrointestinal (GI) absorption, Blood-Brain Barrier (BBB) permeability, and **Lipinski's Rule of Five** (molecular weight, lipophilicity, hydrogen bond givers/receivers).
- **Guided Practice:** Input a chemical structure (SMILES string or compound name) into **SwissADME** and model how to interpret the pharmacokinetic dashboard.
- **Independent Group Work:** Students run their top-performing SwissDock compounds through SwissADME. They critically assess whether their structurally "perfect" binder would actually function as an effective pill or injection inside a living organism, explicitly checking for rule violations or delivery barriers.
- For example: Aspirin and ATB-346 drug SwissADME panels (see images 5 and 6). A SwissADME Interpretation Reference Guide is provided.

Molecule 1



SMILES CC(=O)Oc1ccccc1C(=O)O

Physicochemical Properties	
Formula	C9H8O4
Molecular weight	180.16 g/mol
Num. heavy atoms	13
Num. arom. heavy atoms	6
Fraction Csp3	0.11
Num. rotatable bonds	3
Num. H-bond acceptors	4
Num. H-bond donors	1
Molar Refractivity	44.90
TPSA	63.60 Å²
Lipophilicity	
Log $P_{o/w}$ (iLOGP)	Temporarily unavailable
Log $P_{o/w}$ (XLOGP3)	1.19
Log $P_{o/w}$ (WLOGP)	1.31
Log $P_{o/w}$ (MLOGP)	1.51
Log $P_{o/w}$ (SILICOS-IT)	1.10
Consensus Log $P_{o/w}$	1.28

Water Solubility	
Log *S* (ESOL)	-1.85
Solubility	2.54e+00 mg/ml ; 1.41e-02 mol/l
Class	Very soluble
Log *S* (Ali)	-2.12
Solubility	1.36e+00 mg/ml ; 7.56e-03 mol/l
Class	Soluble
Log *S* (SILICOS-IT)	-1.85
Solubility	2.57e+00 mg/ml ; 1.43e-02 mol/l
Class	Soluble
Pharmacokinetics	
GI absorption	High
BBB permeant	Yes
P-gp substrate	No
CYP1A2 inhibitor	Temporarily unavailable
CYP2C19 inhibitor	Temporarily unavailable
CYP2C9 inhibitor	Temporarily unavailable
CYP2D6 inhibitor	Temporarily unavailable
CYP3A4 inhibitor	Temporarily unavailable
Log K_p (skin permeation)	-6.55 cm/s
Druglikeness	
Lipinski	Yes; 0 violation
Ghose	Yes
Veber	Yes
Egan	Yes
Muegge	No; 1 violation: MW<200
Bioavailability Score	0.85
Medicinal Chemistry	
PAINS	0 alert
Brenk	1 alert: phenol_ester
Leadlikeness	No; 1 violation: MW<250
Synthetic accessibility	1.52

Image 5. SwissADME Panel on Aspirin

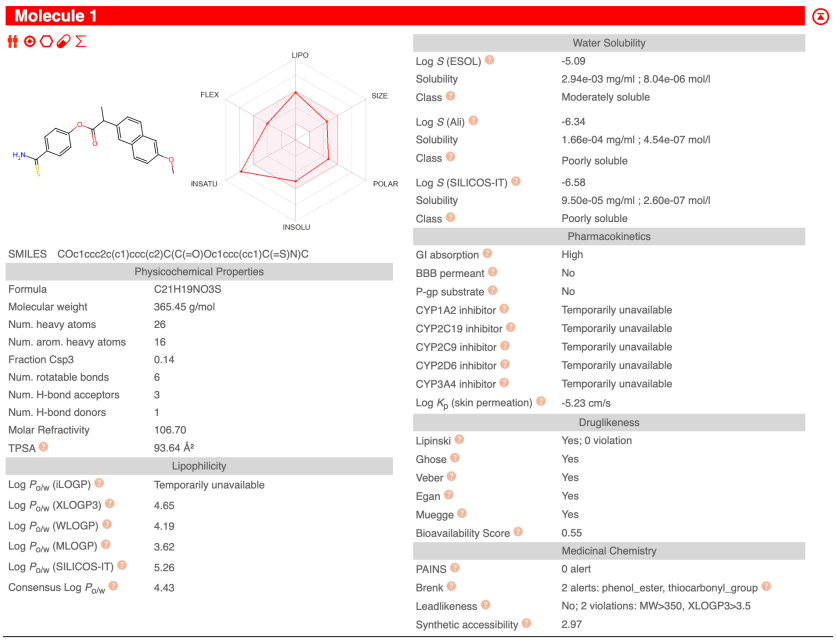


Image 6. SwissADME Panel on ATB-346 Drug

Phase 4: Synthesis, Slide Design & Presentations (Approx. 1-2 Days)

- **The Presentation:** Groups deliver their professional presentations and defend their drug-design recommendations to the class, using the established rubric metrics to evaluate their outcomes.

NGSS Alignment

1. Performance Expectations (PEs)

- **HS-LS1-1:** Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins, which carry out the essential functions of life through systems of specialized cells.
    - *Connection:* Students analyze how specific 3D protein structures (retrieved from the PDB blueprint) govern crucial cellular functions and model how introducing an inhibitor alters those functions to change disease outcomes.
- **HS-LS1-2:** Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.
    - *Connection:* Phase 3 (ADME profiling) directly bridges micro-scale chemical interactions (protein binding) with macro-scale organ systems (Gastrointestinal absorption and Blood-Brain Barrier permeability).
- **HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.
    - *Connection:* Students act as biophysicists optimizing a drug candidate. They break down the massive challenge of curing a disease into distinct computational stages: target identification, thermodynamic binding optimization, and pharmacokinetic delivery profiling.

2. Science and Engineering Practices (SEPs)

- **Developing and Using Models:** Students utilize advanced computational tools (**SwissDock** and **SwissADME**) as predictive digital models to simulate molecular binding and physiological behavior *in silico*.
- **Analyzing and Interpreting Data:** Students systematically evaluate quantitative datasets, specifically using Gibbs Free Energy (Delta G kcal/mol) to evaluate, rank, and compare the efficacy of different chemical ligands against a designated control.

- **Obtaining, Evaluating, and Communicating Information:** The project culminates in a professional symposium-style presentation where students must translate complex bioinformatics data into a coherent, evidence-based scientific argument for their drug's real-world viability.

3. Disciplinary Core Ideas (DCIs)

- **LS1.A: Structure and Function:** All proteins have specific shapes that allow them to interact with other molecules. The binding pocket's surface topography dictates what chemical ligands can fit into it.
- **ETS1.B: Developing Possible Solutions:** High-tech engineering requires optimization. Students test multiple chemical compounds to find the one that balances maximum therapeutic binding (Delta G) with minimal delivery side effects (Lipinski violations).

4. Crosscutting Concepts (CCCs)

- **Structure and Function:** The cornerstone of the entire capstone. Students directly observe how atomic-level structural alignment determines the functional success or failure of a potential therapeutic agent.
- **Systems and System Models:** Students model a chemical compound not as an isolated entity, but as part of an interacting system inside the human body—tracking its path from chemical binding affinity to systemic absorption and boundary permeability.